Mujahid Ali Quidwai

AI Systems Engineer | Agents, Model Post-Training & Evaluation
quidwaiali@gmail.com | LinkedIn | Google Scholar | Portfolio | GitHub

Experience

Kinetic Systems and Technologies San Francisco, CA
Member of Technical Staff (Founding Engineer) *Jun 2026 – Present*

- I build healthcare RL environments and agent evaluations. My work includes reconstructing software from workflow videos and designing RL tasks in the Harbor framework that test how agents interact with it.

Icahn School of Medicine at Mount Sinai New York, NY
Computational Scientist, Oncological Sciences *Dec 2024 – Jun 2026*

- **SHAJAM:** Designed and built a clinical research agent platform spanning records, notes, pathology, imaging, and genomics, with answers traceable to source data and SQL. Connected a source warehouse covering **12M+ patients**; deployed a myeloma instance covering **11,030 patients**.
- **OncoCITE:** Built a multimodal, six-agent system with 22 MCP tools for oncology evidence extraction, source verification, ontology resolution, and resumable runs. In 10 retrospective papers, recovered **21/25 reference items**; experts confirmed **44/45 novel extractions** and found discrepancies in 8/33 curated items. Five prospective papers were evaluated separately. Public preprint and code; under review at Nature Communications.
- **OncoDIF model adaptation:** Fine-tuned a Llama-3 model with LoRA on **21,000 synthetic question–SQL pairs**. Correct SQL improved from **55% to 83.24%** on the study's unseen questions. Work reported in an ASH 2025 conference abstract; evaluated terminology variation and flagged low-confidence outputs for review.
- **GPU genomics:** Designed a 57-process Nextflow DSL2 pipeline for RNA-seq, whole-exome sequencing, and multi-omics integration using NVIDIA Parabricks and RAPIDS. Reduced the observed per-patient workflow from **7 days to 3 hours (56×)**.
- **Collaborative research:** Contributed to myeloma studies on survival modeling, tumor/immune multi-omics integration, and single-cell analysis, connecting computational workflows with scientific questions.

Associate Computational Scientist, Oncological Sciences *Oct 2023 – Dec 2024*

- **Clinical RAG:** Built a retrieval-augmented chatbot for myeloma research with source-linked responses and abstention when retrieval lacked relevant evidence. On the reported oncologist-curated benchmark, achieved similar accuracy and relevance to GPT-3.5/GPT-4 at about **one-tenth the cost** (medRxiv preprint and IMS 2024 conference abstract).
- **Wearables and ML:** Contributed data analysis and predictive modeling to a CAR-T pilot study with 25 evaluable patients, published in JCI Insight in 2026. Wearable monitoring detected 18/20 CRS episodes a median of about seven hours before nursing detection; this was a separate analysis from the predictive models.
- **Domain adaptation:** Fine-tuned Whisper Small for myeloma terminology recognition, with on-device use as the target.
- **Research software:** Built Python REST APIs connecting Epic EHR, REDCap laboratory data, and Nextflow genomics outputs.
- **Mentorship:** Mentored six Carnegie Mellon master's students on CAR-T monitoring research (Sep–Dec 2024), guiding experiment design, baseline comparisons, and reporting in collaboration with the research team.

Independent Engineering & Recognition

AYA – AI for Small Businesses, *NVIDIA Spark Hack NYC* New York, NY · *Apr 2026*

- **Winner, NVIDIA Spark Hack NYC 2026.** Built an agent application for Manhattan small businesses, joining **385 NYC Open Data sources** into business-specific briefs for operational analysis. Public code and demo walkthrough.
- Integrated a React interface, Python data layer, and OpenClaw agent workspaces; dispatched risk, foot-traffic, and competitor analysis to three parallel local model endpoints on the hackathon's NVIDIA GB10-based infrastructure.
- Invited by NVIDIA to present the project at its developer community winners livestream, May 21, 2026.

Earlier Experience

AYA (NYU Entrepreneurial Institute), *Entrepreneurial Fellow* New York, NY · *Jul 2023 – Oct 2023*

- Built AYA, an end-to-end product using a reinforcement-learning-optimized NLP model to extract and generate questions from academic video content; selected for NYU's Summer Sprint (10 of 150 startups).

Growsquares, *Summer Data Scientist* New York, NY · *Jun 2022 – Aug 2022*

- Developed a dynamic growth model predicting plant harvest rates from genotype, weather, and environment; built an ETL pipeline from agricultural journals and weather stations; led the front end for 100+ pilot customers.

Carcrew Technology, *Technical Lead (Founding Team)* New Delhi, India · *Aug 2016 – Sep 2017*

- Built the MVP of an automotive repair marketplace (Django, Flask, PostgreSQL) and led 3 developers on a backend holding 10M+ part-inventory records; scaled to 10,000+ daily active users and supported technical due diligence for a $2M Series A.

Publications & Presentations

Google Scholar: full list

- **OncoCITE: Multimodal Multi-Agent Reconstruction of Clinical Oncology Knowledge Bases from Scientific Literature.** First author. Research Square preprint, 2026; under review at *Nature Communications*.
- **OncoDIF: An auditable AI framework for automated genomic curation and natural-language clinical querying in hematologic malignancies.** First author. *Blood* 146(Suppl 1):2646, 2025. ASH conference abstract.
- **Beyond Black Box AI generated Plagiarism Detection: From Sentence to Document Level.** First author (published as Ali Quidwai), with Chunhui Li and Parijat Dube. BEA 2023 workshop, ACL Anthology, pp. 727–735.
- **A RAG Chatbot for Precision Medicine of Multiple Myeloma.** First author. medRxiv preprint, 2024.
- **Innovative AI-Driven Decision Support Tool for Multiple Myeloma Using Retrieval Augmented Generation.** First author. *Clinical Lymphoma Myeloma and Leukemia* 24:S123–S124, 2024. IMS conference abstract.
- **Detection of cytokine release syndrome using wearables and cytokine profiling following CAR-T therapy for myeloma.** Coauthor; data analysis. *JCI Insight* 11(12):e203988, 2026. Journal article.
- **Multi-Modal Single-Cell Profiling of MRD-Negative Patients Identifies Early Immune Signatures Predicting Prolonged Response or Relapse in Multiple Myeloma.** Coauthor. *Clinical Lymphoma Myeloma and Leukemia* 26:S223, 2026. Conference abstract.
- **Development of predictive relapse indicators for myeloma T cell engagers (PRIME) model for myeloma patients receiving BCMA- and GPRC5D-targeting T cell engagers.** Coauthor. *Blood* 146(Suppl 1):3996, 2025. ASH conference abstract.
- **Integrating Microenvironment with Tumor Multi-Omic using Unsupervised Machine Learning to Model Heterogeneity Refines Multiple Myeloma Subtypes and Reveals Immune-Based Clusters with Prognostic Impact.** Coauthor. *Clinical Lymphoma Myeloma and Leukemia* 25:S181–S182, 2025. IMS conference abstract.

Education

M.S. in Computer Engineering (Honors), NYU Tandon School of Engineering 2021 – 2023

- **Research (first author, ACL 2023 BEA workshop, with IBM Research):** Trained an AI-generated-text detector (contrastive loss, LLM-generated paraphrases) with sentence- and document-level scoring; reached **94% document-level accuracy on HC3**.
- **Teaching Assistant:** Data Structures & Algorithms for Bioinformatics (Prof. Katari), Spring 2022–Spring 2023: taught algorithms and genomic algorithms to 130+ master's students, created assignments, and held office hours.

B.Tech. in Computer Science, Dr. A.P.J. Abdul Kalam Technical University, Ghaziabad, India 2012 – 2016

Technical Skills

Post-training & machine learning: Python, PyTorch, Hugging Face Transformers/PEFT, TRL, Unsloth, LLaMA-Factory, Tinker, Fireworks AI; LoRA/QLoRA SFT, DPO, GRPO, PPO, reward modeling, RLHF; scikit-learn, XGBoost, SHAP, Whisper.

Agent systems & inference: Claude Agent SDK, OpenAI Agents SDK, LangGraph, LangChain, Hermes Agent; MCP/FastMCP, LiteLLM, Pydantic, FastAPI, React, Open WebUI; SGLang/sglang-router, vLLM, Ollama; NVFP4 (ModelOpt), FP8, NVIDIA B200/H100/L40S, NVLink; RAG, hybrid retrieval, Qdrant, Qwen3-Embedding/Reranker-8B, OpenSearch.

Evaluation & observability: Harbor RL environments (containerized tasks, verifiers, reward functions), Inspect; MLflow, LLM-as-judge, Langfuse, OpenTelemetry, OpenAI Evals, DeepEval; source attribution, baseline comparisons, failure analysis.

Computational biology: Nextflow DSL2, Parabricks, DeepVariant, GATK4, BWA-MEM, STAR, fastp, featureCounts, Manta, FACETS, Arriba, PhyloWGS, RAPIDS cuDF, Scanpy, Geneformer.

Clinical data & statistics: OMOP CDM v5.4, SAP HANA, DuckDB, Polars, Parquet, SQL; Epic, REDCap, DICOM, CIViC, ClinVar; pandas, SciPy, statsmodels, lifelines, survival analysis, Plotly.

Cloud platforms: AWS (EC2, S3, ALB/WAF, Lambda, Step Functions, DynamoDB, RDS, SageMaker, CloudWatch), Terraform, Azure OpenAI, Amazon Bedrock; Argo Workflows, Kubernetes, GitHub Actions, CI/CD, Docker, Singularity, LSF, SLURM.